EIN 4343 Inventory and Supply Chain Systems
Spring 26: Homework Assignment 5

1. **[8 points]** WeMove Inc. is a shipping company that provides 3PL service for BWAH Corp. WeMove makes deliveries from BWAH's manufacturing plant to its central warehouse. Based on WeMove's past performance, the transportation lead time for deliveries can assumed to be normally distributed with a mean of 5 days and a standard deviation of 2 days. Currently, WeMove charges BWAH $0.50/unit for shipping costs. Following information is also available:

Average demand at the warehouse	150 units/day
Standard deviation of demand at the warehouse	20 units/day
Fixer ordering cost	$250/order
Unit cost	$40/unit
Shipping quantity	As per EOQ
In-transit inventory carrying rate	15% per year
On-site inventory carrying rate	20% per year
Probability of not stocking out during lead time	90%
Operating days at the warehouse	360 days/year
Out-of-stock costs for the warehouse	Unknown

a. **[2 point]** What are the parameters of demand during lead time?

Average demand during lead time	$\mu_L = (5)(150) = 750$ units
Std Dev of demand during lead time	$\sigma_L = \sqrt{(5)(20^2) + (150^2)(2^2)} = 303.32$ units

b. [3 points] What is the total annual logistics cost (includi4ng on-site inventory, ordering, transportation and in-transit inventory holding) associated with the current system?

Type I service level	$\alpha = F(r) = 0.90 \Rightarrow z = 1.2816$
Safet stock quantity	$s = (303.32)(1.2816) = 388.73$ units
Order quantity	$q = \sqrt{\frac{(2)(250)(150)(360)}{(0.2)(40)}} = 1{,}837.12$ units
Transportation cost	$(0.50)(150)(360) = \$27{,}000/\text{year}$
Ordering cost	$(250)\frac{(150)(360)}{(1{,}837.12)} = \$7{,}348.45/\text{year}$
On-site inventory holding cost	$(0.20)(40)\left(388.73 + \frac{1{,}837.12}{2}\right) = \$10{,}458.32/\text{year}$
In-transit inventory holding cost	$(0.15)(40)(150)(360)\left(\frac{5}{360}\right) = \$4{,}500/\text{year}$
Total logistics cost	$\$49{,}306.71/\text{year}$

c. [3 points] Another company Fast Horses, Inc. has made an offer to BWAH so that the guaranteed transportation lead time with a mean of 4 days and a standard deviation of 1 day, the distribution for which can assumed to be approximately normally distributed. Fast Horses would charge BWAH $1.00/unit for shipping costs. Should BWAH accept this new offer?

Average demand during lead time	$\mu_L^{FH} = (4)(150) = 600$ units
Std Dev of demand during lead time	$\sigma_L^{FH} = \sqrt{(4)(20^2) + (150^2)(1^2)} = 155.24$ units
Type I service level	$\alpha = F(r) = 0.96 \Rightarrow z = 1.2816$
Safet stock quantity	$s = (155.24)(1.2816) = 198.96$ units
Order quantity	$q = \sqrt{\frac{(2)(250)(150)(360)}{(0.2)(40)}} = 1{,}837.12$ units
Transportation cost	$(1)(150)(360) = \$54{,}000/\text{year}$
On-site inventory holding cost	$(0.20)(40)\left(198.96 + \frac{1{,}837.12}{2}\right) = \$8{,}940.16/\text{year}$
In-transit inventory holding cost	$(0.15)(40)(150)(360)\left(\frac{4}{360}\right) = \$3{,}600/\text{year}$
Total logistics cost	$\$73{,}888.63/\text{year}$

BWAH should not accept this offer since the increase in the transportation cost is not covered by the reduction in in-transit inventory holding and on-site inventory holding costs.

2. **[6 points]** Stryker is a manufacturer of surgical medical devices used in knee and hip replacement surgeries. Currently, these devices are distributed to hospitals around the nation via Stryker's 18 regional distribution centers. The weekly demand for these medical devices at a distribution center appears to follow an approximately normal distribution with a mean of 645 devices and a standard deviation of 48 devices. The weekly demand at the distribution centers are independent and are identically distributed. It costs Stryker $50/unit per device. Stryker uses an annual inventory carrying rate of 30% on site, and 15% in transit. Stryker strives towards for a Type I service level of 90%. Stryker outsources the shipment of medical devices from their manufacturing facility in Florida to the 18 distribution centers to Cone Logistics that allows Stryker to choose the shipment quantity and charges Stryker $3,000 per shipment and guarantees a delivery lead time of 2 weeks to anywhere in the nation. Assume that there are 48 weeks in a year. What are the total annual logistics costs for Stryker?

Shipment quantity for one DC	$q_i = \sqrt{\frac{(2)(3{,}000)(48)(645)}{(0.3)(50)}} = 3{,}519.09 \text{ units}$
Average demand during lead time	$\mu_L = (2)(645) = 1{,}290 \text{ units}$
Std Dev of demand during lead time	$\sigma_L = \sqrt{(2)(48^2)} = 67.88 \text{ units}$
Type I service level	$\alpha = F(r) = 0.90 \Rightarrow z = 1.2816$
Safet stock quantity	$s_i = (1.2816)(67.88) = 87.00 \text{ units}$
Transportation cost	$(3{,}000)\frac{(48)(645)}{3{,}519.09} = \$26{,}393.19/\text{year}$
On-site inventory holding cost	$(0.3)(50)\left(87.00 + \frac{3{,}519.09}{2}\right) = \$26{,}860.13/\text{year}$
In-transit inventory holding cost	$(0.15)(50)(48)(645)\left(\frac{2}{48}\right) = \$9{,}675/\text{year}$
Total cost for one DC	$\$62{,}928.32/\text{year}$
Total cost for the entire DC network	$(18)(35{,}860.79) = \$1{,}132{,}709.76/\text{year}$

3. **[6 points]** Consider the data in Question 2. Suppose now that a third-party logistics provider has offered to take over distribution operations of Stryker. Specifically, they offer to operate a single warehouse in Macon, GA from where the devices are to be shipped directly to hospitals via overnight delivery. The warehouse in Macon, GA, will be replenished once a week from Stryker's facility in Florida, and the delivery lead time would be less than 6 hours and hence, in-transit inventory holding costs would be negligible. Suppose that an average hospital order is 15 devices, and the third-party logistics company will charge Stryker $80 per direct hospital delivery and will also charge Stryker a flat fee of $30,000 per year for warehouse space. Should Stryker accept this offer?

Order quantity for the warehouse	$q = (18)(645) = 11{,}610 \text{ units}$
Transportation cost	$(80)\dfrac{(48)(645)(18)}{15} = \$2{,}972{,}160/\text{year}$
Annual warehousing cost	$\$30{,}000/\text{year}$
In-transit inventory holding cost	$\$0/\text{year}$
On-site inventory holding cost	$(0.3)(50)\left(\dfrac{11{,}610}{2}\right) = \$87{,}075/\text{year}$
Total cost	$\$3{,}089{,}235/\text{year}$

Stryker should not accept this offer. The transportation cost per shipment at $80/shipment appears to be too high. If it were $25 per shipment, it would have been preferrable. Hence, a sensitivity analysis can be conducted on certain cost parameters to determine when an option be preferrable which may be quite useful in contract negotiations.